

Hajar Mrifag

Software Engineering Student | Backend & Full-stack | Class of 2027

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EXPERIENCE

Data & AI Engineering Intern | March - April 2026

Chengdu Suncaper Data Co., Ltd | SkyQuery flight analytics platform

- Designed the HDFS/Hive data model and PySpark pipelines in a two-person team, enabling querying and analysis across more than 15 million flight records.
- Developed a FastAPI text-to-SQL prototype with schema context, query-prefix validation and a three-attempt correction loop; the published service executes against SQLite.
- Deployed the backend to AWS EC2 through GitHub Actions; added structured logging and a metrics dashboard to inspect request failures and system behaviour during testing.
- Delivered 14 Jira stories across three Agile sprints and presented the platform and analysis findings to the team.

EDUCATION

Sichuan University | Software Engineering | Expected graduation: 2027

TECHNICAL SKILLS

Languages: Python, C#, TypeScript, SQL, C++ / CUDA

Applications: FastAPI, ASP.NET Core, React, PostgreSQL

Delivery & data: Docker, GitHub Actions, Git, PySpark, Hive, HDFS

SELECTED ENGINEERING PROJECTS

GridFlex | React, TypeScript, FastAPI, NumPy

- Built and deployed an energy-simulation workspace with a React client, FastAPI service and numerical engine; release 0.2.0 includes 91 Python tests and five browser journeys.
- Reduced the shift + battery pipeline from 71.731 to 2.246 ms (31.9x) on a recorded local 60-day benchmark; preserved numerical outputs for that case.

Aegis Claims | ASP.NET Core, PostgreSQL, React

- Developed a claims application with JWT authentication, API-enforced roles, document signature validation and an audit trail for status changes.
- Documented threat scenarios and deployment limits; added unit, integration and browser tests with automated security checks.

Neobank API & Analytics | FastAPI, PostgreSQL, SQL, Docker

- Implemented atomic transfers, idempotent retries and consistently ordered account locks, with database-backed concurrency and rollback tests.
- Built reproducible analytics over 891,772 synthetic transactions, separating statistical uplift from incentive profitability.

CUDA Transformer Kernels | CUDA C++, Nsight Compute

- Implemented and profiled GEMM and Softmax kernels with CPU-reference checks. Register-tiled FP32 GEMM reached 1.28 TFLOPS on a T4 at 1024 x 1024, 3.32x the naive kernel and 24% of cuBLAS (median of 20 runs).